Satyam Sangeet *PhD Candidate* (*Founder - CompObelisk*)

India

satyam_verma20@yahoo.in

(+91)-7028510659

LinkedIn

Research Gate

Company Website

Github

ORCID

Google Scholar

Personal Page

Education

2024/01 – present
Sydney, Australia

Doctor of Philosophy (Chronophysics & Chronobiology)
University of Sydney

2018/07 – 2020/06
Bhopal, India

Master of Technology (Biotechnology)
Maulana Azad National Institute of Technology

- Average GPA - 9.59/10
- Master Dissertation - "**Effect of Stochasticity on neuronal signaling**" under the guidance of Dr. Suhita Nadkarni at Indian Institute of Science Education and Research (IISER) - Pune (12 months)

2014/08 – 2018/05
Pune, India

Bachelor of Technology (Biotechnology)
Dr D.Y. Patil Biotechnology and Bioinformatics Institute

- Average GPA - 9.42/10
- Bachelor Dissertation - "**Development of Microbial zombies using synthesized metal nanoparticles**" under the guidance of Dr. Swapnil Gaikwad at Dr. D.Y. Patil Biotechnology & Bioinformatics Institute (6 months)
- Sports Secretary

Awards

2026

Postgraduate Research Support Scheme (PRSS) Travel Grant
University of Sydney
Awarded the PRSS Travel grant for the European Sleep Research Society (ESRS) in Maastricht, Netherlands

2025

Postgraduate Research Support Scheme (PRSS) Travel Grant
University of Sydney
Awarded the PRSS Travel grant for the World Sleep Conference in Singapore (September 2025)

2024

Faculty of Science Research Scholarship
University of Sydney
Awarded the Faculty of Science Research Scholarship for pursuing PhD in the field of Computational Neuroscience

2024

Centre for Chronic Diseases of Ageing (CCDA)
Woolcock Institute of Medical Research
Awarded the top up scholarship from Woolcock Institute of Medical Research to pursue my PhD

2021/01/05 **Har Gobind Khorana IYBA (Innovative Young Biotechnologist Award) Junior Research Fellow Award**

Department of Biotechnology (*DBT*), *Government of India*

Awarded the prestigious Har Gobind Khorana IYBA Junior Research Fellow Award for the project funded by the Department of Biotechnology (DBT), Government of India, to be conducted at Indian Institute of Science Education and Research (IISER) - Kolkata

2020/10/12 **Junior Research Fellow**

Department of Biotechnology (*DBT*), *Government of India*

Awarded the Junior Research Fellow Award for the project funded by the Department of Biotechnology (DBT), Government of India, to be conducted at Indian Institute of Science Education and Research (IISER) - Bhopal

2018/07/19 **GATE Scholarship**

AICTE, Government of India

Awarded the GATE scholarship by AICTE, Government of India for the completion of Master studies

2018 **Subroto Memorial Scholarship**

Indian Air Force Benevolent Association

Awarded the Subroto Memorial Scholarship for academic excellence in Bachelor Studies by Indian Air Force.

2013 **Subroto Memorial Scholarship**

Indian Air Force Benevolent Association

Awarded the Subroto Memorial Scholarship for academic excellence in Middle and High school

2011 **Jagdish Bose National Science Talent Search**

Department of Science and Technology (*DST*), *Government of India*

Represented my high school at JBNSTS organized by the Department of Science and Technology (DST), Government of India

Independent Technical Projects

EVOLVE

Developed a web-server that allows an easy user interface to perform feature calculation and machine learning model training on viral protein sequences to predict future mutations in variants of the virus. This webserver is developed based on my previous published work: "***Quantifying Mutational Response to Track the Evolution of SARS-CoV-2 Spike Variants: Introducing a Statistical-Mechanics-Guided Machine Learning Method*** "

SIMulation ANAlysis (SIMANA)

Developed SIMANA, a webtool **for analyzing and visualizing molecular dynamics simulation data** alongwith performing multiple drug-discovery related calculations. The platform aims to streamline the analysis process and enable deeper insights into molecular behavior.

ARIAS (Article Research Intelligence And Summarization)

Developed a webserver that allows easy summarization of the research articles. The webserver also allows the user to interact with the research article and ask specific questions based on the published work. I integrated the language model ChatGPT to make the summarization and questions asked by the user to be interactive with the AI.

Research Experience

2021/01 – 2023/12 **Tracking Mutational Response of SARS-CoV-2 Variants**
Har Gobind Khorana Junior Research Fellow | *Indian Institute of Science Education and Research* (*Kolkata*) | *PI*: *Dr. Susmita Roy*

- Planned and organized the experiments to investigate the genomic fluctuations of SARS-CoV-2 sequences
- Visualized the mutational pattern of different Variant of Concerns (VOCs) and Variant of Interests (VOIs)
- Developed mathematical parameter and multi-layer perceptron model to quantify the mutational response

2020/10 – 2020/12 **Synthesis of Double Knot Toxins (DkTx) and sub-variants for TRPV1 Activation**
Junior Research Fellow | *Indian Institute of Science Education and Research* (*Bhopal*) | *PI*: *Dr. Jeet Kalia*

- Reviewed pieces of literature on Double knot toxins (DkTx) and their role in ion channel activation and concluded that the development of hybrid variants of DkTx will allow investigating the mechanism of ion channel activation.
- Purified DkTx and different variants using HPLC and buffer preparations.
- Handled the culture of *Xenopus laevis*

2019/05 – 2020/06 **Effect of stochasticity on Neuronal Signalling**
M.Tech Dissertation | *Indian Institute of Science Education and Research* (*Pune*) | *PI*: *Dr. Suhita Nadkarni*

- Constructed the neural signaling model of CA3-CA1 hippocampal synapse.
- Inspected the effect of noise on neural signaling by varying ion channel numbers.

2019/01 – 2019/04 **Targeting Chemotaxis Inhibitory Proteins (CHIP) of Staphylococcus aureus using bio-inspired Iron nanoparticles**
Independent Research Project | *Maulana Azad National Institute of Technology* | *PI*: *Dr. Khushhali M. Pandey*

- Developed iron nanoparticles using plant source against Staphylococcus aureus.
- Determined the surface topology of the nanoparticles by employing different characterization techniques.
- Mentored 10 undergraduate students for their bachelor thesis.
- Addressed the results in a reputed international conference.

2018/01 – 2018/05 **Development of Microbial Zombies using synthesised metal nanoparticles**
B.Tech Dissertation | *Dr. D.Y. Patil Biotechnology & Bioinformatics Institute* | *PI*: *Dr. Swapnil Gaikwad*

- Developed and characterized silver nanoparticles to examine their antibacterial activity against *Pseudomonas aerugionsa*.
- Systematized the protocol for nanoparticle synthesis and reported the observations to a reputed journal

Publications

A readthrough-permissive stop context and C-terminal constraint distinguish AQP4ex from its orthodox aquaporin paralogues
Research Square (2026)
Sangeet S

doi: 10.21203/rs.3.rs-11323608/v1

Quantitative Modelling of Amyloid-β Dynamics in Brain, CSF, and Plasma During Sleep and Wakefulness
BioRxiv (2026)
Sangeet S, Phillips CL, Hoyos CM, D'Rozario AL, Lucey BP, Postnova S
doi: 10.64898/2026.09.07.749991

Modelling Amyloid-beta clearance dynamics across sleep-wake cycles in healthy conditions
World Sleep Congress, Sleep Medicine (2026)
Sangeet S, Phillips C, Hoyos C, D'Rozario A, Postnova S
doi: Article 106992

EVOLVE: A Web Platform for AI-Based Protein Mutation Prediction and Evolutionary Phase Exploration
Journal of Chemical Information and Modelling
Sangeet S, Sinha A, Nair MB, Mahata A, Sarkar R, Roy S
doi: 10.1021/acs.jcim.5c00026

Bacopa monnieri phytochemicals as promising BACE1 inhibitors for Alzheimer's disease therapy
Scientific Reports
Sangeet S, Khan A
doi: 10.1038/s41598-025-92644-y

Spectroscopic and computational study of molecular interaction of Pexidartinib with homologous mammalian transport proteins
Journal of Molecular Liquids
Ankita Jalan, Amit Kumar Pradhan, **Satyam Sangeet**, N Shaemningwar Moyon
doi: 10.1016/j.molliq.2024.125869

RNA's Dynamic Conformational Selection and Entropic Allosteric Mechanism in Controlling Cascade Protein Binding Event
Journal of Physical Chemistry Letters (2024)
Amrita Chakraborty, Dibyamanjaree Samant, Raju Sarkar, **Satyam Sangeet**, Sangram Prusty, Susmita Roy
doi: 10.1021/acs.jpclett.4c00740

Exploring the interaction of a potent anti-cancer drug Selumetinib with bovine serum albumin: Spectral and computational attributes
Journal of Molecular Recognition (2024)
Jalan A, **Sangeet S**, Pradhan AK, Moyon NS
doi: 10.1002/jmr.3084

An in-silico approach to identify bioactive phytochemicals from Houttuynia cordata Thunb. As potential inhibitors of human glutathione reductase
Journal of Biomolecular Structure and Dynamics (2023)
Sangeet S, Khan A
doi: 10.1080/07391102.2023.2294181

Evolution of Sequence and Structure of SARS-CoV-2 Spike Protein: A Dynamic Perspective
ACS Omega (2023)
Sinha A*, **Sangeet S***, Roy S (*equal authorship)
doi: 10.1021/acsomega.3c00944

The quest for novel drugs against COVID-19: In silico study
Tissue Engineering and Regenerative Medicine, CRC Press
Prajapati S, Zeenat L, **Sangeet S**, Khan A, Pandey KM
doi: 10.1201/9781003536352-18

In-silico screening of potential phytochemicals against Extracellular Adherence Proteins (EAP) of Staphylococcus aureus from Indian Medicinal Plants
Research Journal of Pharmacy and Technology (2023)
Zeenat L, Prajapati S, **Sangeet S**, Khan A, Pandey KM
doi: 10.52711/0974-360X.2023.00762

Quantifying Mutational Response to track the evolution of SARS-CoV-2 Spike variants: Introducing a Statistical Mechanics-guided Machine Learning Method
The Journal of Physical Chemistry B (2022)
Sangeet S, Sarkar R, Mohanty SK, Roy S
doi: 10.1021/*acs.jpcb.2c*04574

Computational Analysis of Bacopa monnieri (L.) Wettst. compounds for drug development against Neurodegenerative Disorders
Current Computer Aided Drug Design (2022)
Sangeet S, Khan A, Mahanta S, Roy N, Das SK, Mohanta YK, Saravanan M, Tag H, Hui PK
doi: 10.2174/1573409918666221010103652

Exploratory Data Analysis of Genomic Sequence of Variants of SARS-CoV-2 Reveals Sequence Divergence and Mutational Localisation
Bioinformatics and Biology Insights (2022)
Sangeet S, Khan A
doi: 10.1177/11779322221126294

Computational Approach to attenuate the virulence of Pseudomonas aeruginosa through Bioinspired silver nanoparticles
3 Biotech, Springer (2022)
Sangeet S, Pawar S, Nawani N, Junarkar M, Gaikwad S
doi: 10.1007/s13205-022-03367-0

Antibacterial property of Biologically synthesized Iron Nanoparticles against Staphylococcus aureus
International Conference on Advancement in Materials, Manufacturing, and Energy Engineering (*ICAMME*-2021). *Chapter* 58, *Springer Nature* (2021)
Sangeet S, Khan A, Prabha S, Menaria K
doi:10.1007/978-981-16-8341-1_7

Drug Development against SARS-CoV-2 from Indian Medicinal Plants - Computational Approach
International Journal of Engineering Applied Sciences and Technology (2020)
Sangeet S, Khan A
doi:10.33564/IJEAST.2020.v05i06.047

Academic Conference and Co-curricular Activities

2026 **European Sleep Research Society**
Maastricht, Netherlands *Poster Presentation*

2025 Singapore
World Sleep Congress
Poster Presentation
Presented my poster titled "***Modelling Amyloid-beta Dynamics Modulated by Sleep-Wake Cycles***" at the World Sleep Conference held at Singapore

2025 Adelaide
Australasian Chronobiology Society
Poster Presentation
Delivered an oral presentation for my work titled "Modelling Amyloid-beta Dynamics Modulated by Sleep-Wake Cycles" at the Australasian Chronobiology Society held in Adelaide

2023 Paris
iGEM IISER Kolkata
Developing Machine Learning Models for Mutation Prediction in Xanthomonas oryzae
Won Gold Medal in iGEM 2023 held in France on the topic "Understanding the Social Language and Virulence Mechanisms of Bacteria: Strategy for Designing Disease Interference for *Xanthomonas oryzae*"

2018 Pune
Volunteered in Industry-Academia Conclave organised by Dr. D.Y. Patil Biotechnology & Bioinformatics Institute

2016 Pune
Usage of DNA as Memory Storage Device
Poster Presentation
Sangeet S, Singh D | Dr. D.Y. Patil Biotechnology and Bioinformatics Institute

Invited Talks

2023/03
Invited as a Speaker/Tutor to deliver a workshop on "Molecular Docking & Molecular Dynamic Simulations for Drug Discovery" for Faculty Members
Part of Faculty Training Program
NEMCARE Group of Institutions

2022/01
Invited as a guest lecturer for Science Foundation on the topic "Exploring DNA using Biopython"
Deen Dayal Upadhyay College, University of Delhi

2021/02
Workshop on "Molecular Docking in Drug Screening"
CompObelisk

2020/08 – 2020/09
Workshop on "Importance of Python & Biopython for Data analysis in Biological Systems"
LifeGenBio

Technical Skills

Microbiological techniques	●●●●○	Analytical Techniques	●●●○○
Computational and Bioinformatics	●●●●○	Molecular Biology	●●●○○
Machine Learning	●●●○○		

Relevant Courses

Computational Biology
Bioinformatics, Cheminformatics, Drug Screening, Structural Biology, Data Analysis, Computational Neuroscience

Programming

Python, Biopython, R, C-programming, Matlab

Molecular Biology & Nanotechnology

Molecular Biology, Analytical Skills, Immunology, Animal Tissue Culture, Biosensor, and Nanotechnology, Virology

Microbiology and Cell BIology

Microbiology, Mammalian Physiology, Culture techniques, Cell Biology

Bioinformatic Skills

BLAST, Genomic Data Analysis, Database Scrapping, Machine Learning with Omics Data, Protein Sequence Analysis, Sequence Alignments